

Supplementary Material: Feature-Aligned Adaptive Triggers for Effective and Stealthy Clean-Label Backdoor Attacks

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1. FAAT training procedure

Algorithm 1 gives the full self-contained training procedure of Sec. 3 of the main paper: the global direction δ_g is optimized first and frozen, the poison set is chosen by the res_x^2 criterion, and the victim and adaptive generator g_ϕ are then trained jointly. It requires only the dataset and a target label.

Algorithm 1 FAAT training (self-contained)

Input: dataset $\mathcal{D}$, target class t , poison rate ρ , budgets

$\tau_{\ell_2}, \tau_{\text{DCT}}$

- 1: Optimize global direction δ_g on a clean proxy; **freeze**.
- 2: Select $\rho|\mathcal{D}_t|$ target-class images via res_x^2 criterion $\rightarrow$ poison set $\mathcal{P}$.
- 3: **for** each epoch **do**
- 4: Sample batch; form $\tilde{x}_i = \text{clip}(x_i + \delta_g + g_\phi(x_i))$ for $i \in \mathcal{P}$, projecting $g_\phi(x_i)$ onto the ℓ_2/DCT budget (Eq. (2) of the main paper).
- 5: Update victim and g_ϕ by SGD on $\mathcal{L}_{\text{CE}}^{\text{clean}} + \lambda_p \mathcal{L}_{\text{CE}}^{\text{poison}} + \lambda_a \mathcal{L}_{\text{align}}$.
- 6: **end for**

Output: victim model (backdoored).

2. Per-dataset FAAT detail

Table 1 gives FAAT’s full per- ℓ_2 , 3-seed results (ASR mean $\pm$ std). All values are end-of-training (last-20-epoch) means. GTSRB uses a larger budget because of its 43-class, BA $\approx$ 100% regime.

3. Defense detail (selection $\times$ defense)

Table 2 reports post-defense ASR and benign accuracy for the Component-A (res_x^2) vs. random selection on the baseline authors’ defense benchmark. The strongest model-repair defenses (ABL, RNP) reduce ASR but also collapse benign accuracy (RNP: BA $\downarrow$ to 44), limiting their practical use.

Dataset (rate)	ℓ_2	ASR	σ_{ASR}	BA	SSIM	SS*
CIFAR-10 (1%)	0.9	76.1	2.5	94.8	0.981	0.438
	1.2	88.6	0.9	94.7	0.967	0.455
	1.5	93.6	1.9	94.8	0.950	0.433
GTSRB (1%)	2.5	64.5	3.4	99.9	0.805	0.790
	3.0	71.5	1.8	99.9	0.761	0.733
	3.5	82.7	2.7	99.8	0.720	0.737
CIFAR-100 (1%)	1.5	99.1	0.1	77.1	0.949	0.208
	2.0	99.6	0.1	77.3	0.919	0.200
	2.5	99.6	0.1	77.0	0.888	0.212
CIFAR-100 (0.5%)	1.5	96.8	0.6	77.9	0.949	0.262
	2.0	98.5	0.6	77.6	0.919	0.246
	2.5	99.4	0.2	77.9	0.888	0.236
Tiny-IN (0.25%)	1.5	93.0	2.0	55.0	0.952	0.265
	2.0	95.8	1.3	54.7	0.927	0.249
	2.5	94.8	0.9	54.5	0.902	0.248

Table 1. FAAT per- ℓ_2 detail (3 seeds). *SS-AUC reported as 1-AUC (lower = more evasive). Source: docs/{v4, v6, v6c, v7}.results.json.

Defense	res_x^2 (ours)		random	
	ASR $\downarrow$	BA	ASR $\downarrow$	BA
AC [1]	75.1	89.2	55.2	89.7
NC [5]	75.5	92.5	42.9	93.0
FST	76.6	91.8	54.0	92.5
FP [4]	66.4	92.2	50.6	92.4
ABL [2]	21.8	62.1	19.2	61.1
RNP [3]	24.8	44.1	12.6	51.3
I-BAU [6]	23.1	87.9	21.4	88.6

Table 2. Post-defense ASR (%) and BA (%) by selection. Source: defense_results/summary.csv (authors’ logs).

4. KST multi-dataset campaign

Table 3 reports KST at the operating $\varepsilon = 48/255$ on the harder datasets (where the flat-spectrum trigger becomes learnable), with the lower- ε GTSRB rows showing the

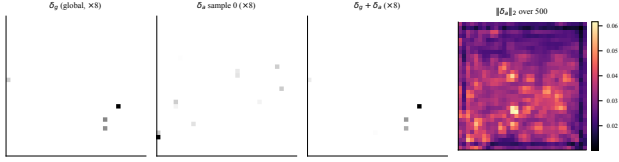


Figure 1. **FAAT trigger visualization** (CIFAR-10). Global δ_g , a sample adaptive residual $\delta_a^{(0)}$, their sum (all amplified $\times 8$), and the average ℓ_2 -norm map of δ_a across 500 poisoned samples.

stealth-but-no-ASR regime.

Dataset (rate)	ε	selection	ASR	BA
CIFAR-100 (0.5%)	48	forget	83.3	78.1
	48	res_x	93.9	78.0
Tiny-IN (0.25%)	48	forget	98.5	55.4
	48	res_x	98.5	55.2
GTSRB (1%)	20	forget / res_x	0.0	100.0
	48	forget	$0.8^\dagger$	100.0

Table 3. KST multi-dataset. † peak 19.3% at epoch 139 then collapse to 0.8% (Fig. 9 of the main paper). On GTSRB KST stays stealthy (SSIM 0.94 at $\varepsilon = 20$) but fails to flip the overconfident victim. Sources: `results/kst_sdt/c100_kst_e48_*/output_1.log`, `tiny_kst_e48_*/output_1.log`, `gtsrb_kst_e48_forget/output_1.log`.

5. Reproduced CIFAR-10 Table-1

Table 4 is our full reproduction of the clean-label components’ Table-1 (4 attacks $\times$ 8 selections, ASR last-20-epoch mean), confirming the selection ranking (res_x and res_x^2 dominate) used to choose Component-A.

selection	BadNets-C	Blended-C	MultiBpp-RGB	MultiBpp-B
random	37.2	53.4	1.2	1.4
loss	52.7	59.4	47.9	28.0
gradient	52.6	60.2	53.3	38.3
forget	71.7	71.1	78.1	81.0
res-log	75.9	82.3	82.1	89.7
res-x	68.8	82.3	85.1	89.7
res-x ²	70.6	75.9	82.5	85.4
res-e ^x	66.7	71.8	62.3	53.9

Table 4. Reproduced CIFAR-10 ASR (%), 1% poison, 300 epochs. Source: `results/_detail_summary.json`.

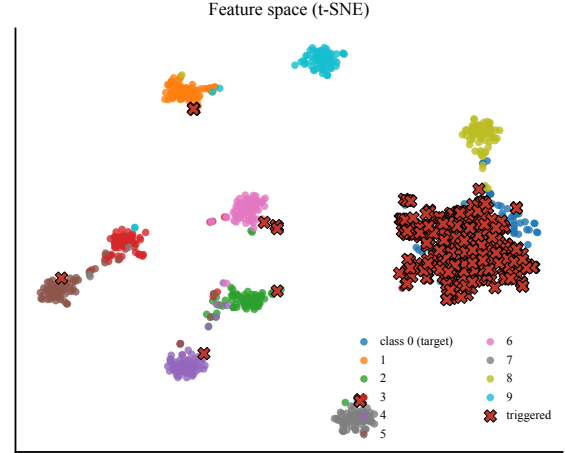


Figure 2. **Feature-space t-SNE** of the FAAT victim (CIFAR-10). Each class is a color; red $\times$ are non-target test images with the test trigger applied. Triggered images migrate into the target-class (class 0) cluster, confirming the global direction is a strong back-door signal in feature space.

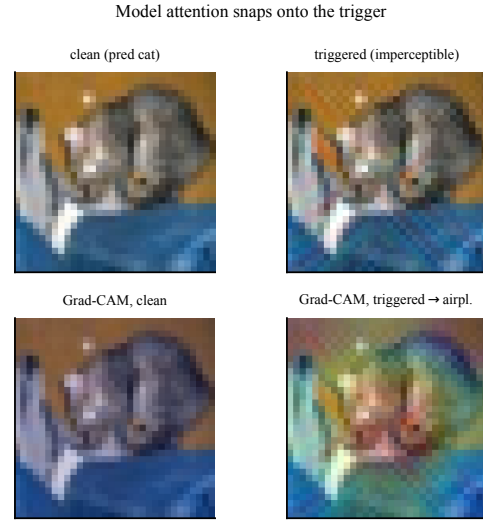


Figure 3. **Grad-CAM on the target logit** (CIFAR-10). On a clean cat image the model attends normally; after the imperceptible trigger the attention snaps onto the trigger and the prediction flips to the target class.

6. Additional visualizations

7. Physical robustness

Figure 5 evaluates the deployed trigger under test-time input transformations. The backdoor tracks benign accuracy under degradation: it is not disproportionately fragile, and severe corruptions that break the attack (e.g. JPEG quality ≤ 30) also destroy benign utility.

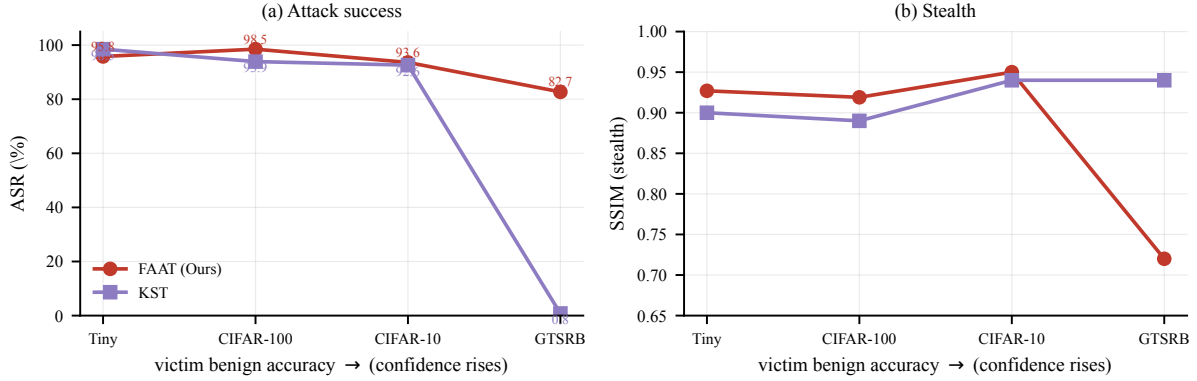


Figure 4. **The ASR–stealth window narrows with victim confidence.** Moving right (Tiny → CIFAR-100 → CIFAR-10 → GTSRB, by rising benign accuracy), FAAT holds ASR but loses stealth on GTSRB, while KST holds stealth but loses ASR. No method keeps both at the high-confidence end.

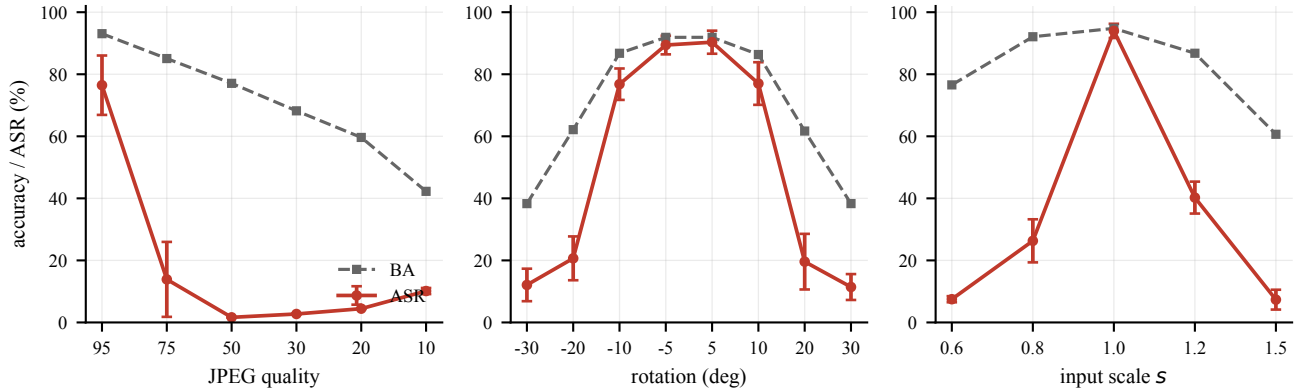


Figure 5. **Physical robustness** of FAAT (CIFAR-10, $\ell_2 = 1.5$, 3 seeds, mean $\pm$ std) under test-time transformations applied to the input before inference: JPEG compression (quality 10–95), rotation (± 5 – 30°), and rescaling ($s = 0.6$ – 1.5 with zero-pad or centre-crop back to 32×32). The backdoor survives moderate perturbations (ASR > 75% up to quality 95, $\pm 5^\circ$, $s = 0.8$) and degrades gracefully alongside benign accuracy under severe ones — the attack is no more fragile than the benign task itself.

References

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- [2] Yige Li, Xixiang Lyu, Nodens Koren, Lingbo Lyu, Bo Li, and Xingjun Ma. Anti-backdoor learning: Training clean models on poisoned data. In *ICLR*, 2021.
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- [5] Bolun Wang, Yuanshun Yao, Shawn Shan, Huiying Li, Bi-mal Viswanath, Haitao Zheng, and Ben Y. Zhao. Neural cleanse: Identifying and mitigating backdoor attacks in neural networks. In *IEEE Symposium on Security and Privacy (S&P)*, 2019.
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